Prajwal Patil

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OBJECTIVE

Aspiring AI/ML Engineer with hands-on experience building RAG applications, LangChain pipelines, and LLM-powered systems. Skilled in Python, data processing, and modern AI frameworks, seeking an opportunity to apply my technical skills to real-world problems and contribute to innovative AI-driven products.

EDUCATION

BE in Computer Science and Engineering | BMS College of Engineering | **CGPA: 8.1** Sep 2024

PUC Science class XI & XII | KLE'S Prerana PU Science College, Hubballi | **Percentage - 94.5%** Apr 2019

SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, Dart
- **Libraries & Frameworks:** Pandas, Matplotlib, Scikit-learn, Seaborn, TensorFlow, Keras, Django, React.js, Flutter, Node.js, Express.js
- **LLM & RAG Ecosystem:** LangChain, Retrieval-Augmented Generation (RAG), Ollama, ChromaDB, HuggingFace Transformers & Embedding Models, Groq LLM APIs
- **Databases:** MySQL, Oracle SQL

CERTIFICATES

- Introduction to Data Science in Python
- Applied Plotting, Charting & Data Representation in Python
- Applied Machine Learning in Python
- Web Application Technologies and Django

PROJECTS

Object Classification and Localization using TensorFlow (Emoji dataset) (GitHub)

Used: **Python, TensorFlow, NumPy, Matplotlib, Keras, CNN, custom metrics, callbacks**

- Built a CNN that simultaneously performs image classification and bounding box regression (predicting emoji position)
- Synthetic Dataset Generation: Created training data by randomly placing emoji images on 144x144 pixel canvases.
- Custom Metric: Implemented Intersection over Union (IoU) as a custom Keras metric to evaluate the accuracy of bounding box predictions during training.

Aircraft Damage Classification & Captioning Using Deep Learning (GitHub)

Used: **Python, TensorFlow, Keras, VGG16, Transformer-based BLIP, Deep Learning, Computer Vision**

- Developed and evaluated a deep learning pipeline for automated aircraft damage detection and captioning.
- Leveraged a pre-trained VGG16 model for classifying aircraft images as either "dent" or "crack," improving efficiency and accuracy over manual inspection.
- Integrated a Transformer-based BLIP model to generate detailed captions and summaries of detected damage.

Haryana Gaushala RAG Search System (GitHub)

Used: **LangChain, Ollama, ChromaDB, RAG, Document Chunks, Llama-3.1**

- Developed a RAG-based search system by extracting PDF tables using pdfplumber, converting them into structured Pandas dataframes, and transforming rows into clean text.
- Implemented recursive text splitting, E5-base-v2 embeddings, and persistent ChromaDB vector storage for efficient semantic retrieval.
- Integrated Groq Llama-3.1-8B-Instant with strict prompt templates to generate accurate, context-grounded responses.

INTERNSHIPS

SEE Room Allocation Web App (Internship Project)

Technologies: ReactJS, Node.js, Express.js

- Designed and implemented a responsive web application to streamline exam room allocation.
- Gained practical experience with RESTful APIs and real-time form handling.

Intern – mSense.ai | Flutter, MQTT, Python, ThingsBoard. **(Jan 2026 to till date)**

- Developed a **Flutter application (**Digital Eyes) for Zebra TC21 devices for production-line monitoring at SMIPL.
- Integrated app with ThingsBoard IoT platform using **MQTT (Mosquitto) and Python** for real-time device communication.
- Designed and customized ThingsBoard widgets for live monitoring dashboards.
- Implemented secure device provisioning and MQTT topic management.